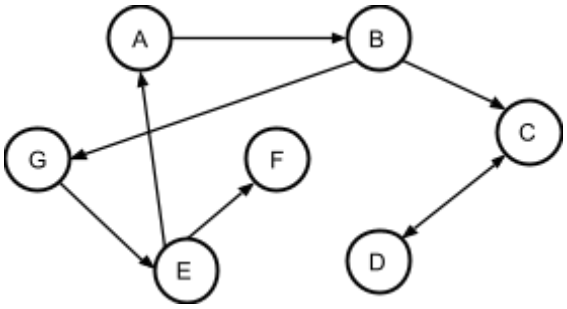
American Computer Science League

2025-2026 ● Contest 4: Short Problems ● Senior Division

1. Graph Theory

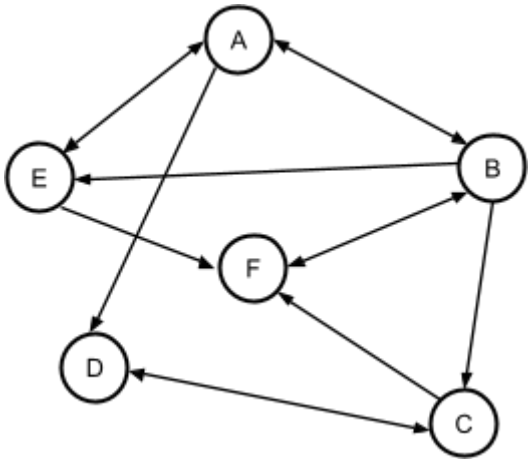
How many ordered pairs of vertices have no paths of length 2 in the following directed graph? For example, (B,A) means the path from B to A.



A. 28
B. 39
C. 40
D. 41
E. 43

2. Graph Theory

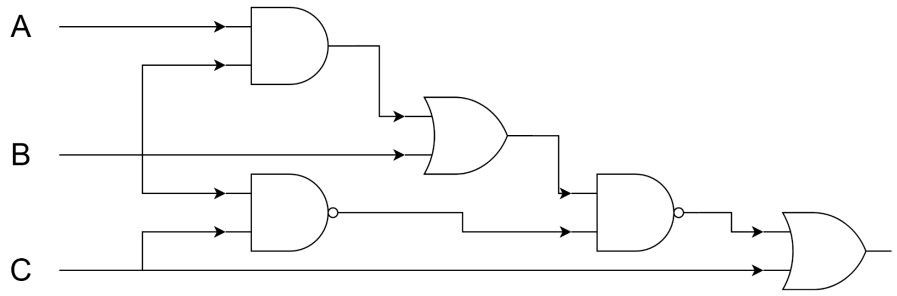
How many cycles containing vertex B are there in this directed graph?



A. 4
B. 5
C. 6
D. 7
E. 8

3. Digital Electronics

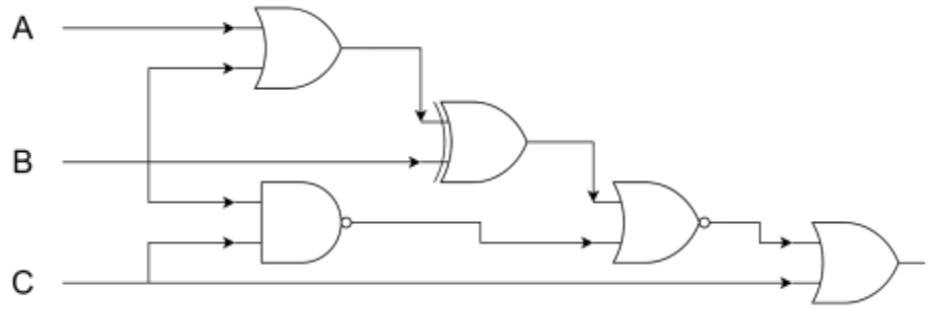
How many ordered triples make the following digital circuit TRUE?



A. 0
B. 2
C. 4
D. 6
E. 8

4. Digital Electronics

Simplify the Boolean algebra expression represented by the following digital circuit using the fewest operations possible:



A. $B\overline{C}$
B. $BC + \overline{C}$
C. C
D. $B + C$
E. $B + \overline{C}$

5. Assembly Language

What is the output when the following program is executed?

```
A       DC      3
B       DC      7
        LOAD    A
        MULT    B
        STORE   P
        DIV     =10
        STORE   C
        MULT    =-10
        ADD     P
        STORE   E
        ADD     C
        MULT    =10
        STORE   D
        LOAD    C
        MULT    =100
        ADD     D
        ADD     E
        STORE   ANS
        PRINT   ANS
        END
```

A. 231
B. 132
C. 321
D. 213
E. 123

6. Assembly Language

What is the output when the following program is executed with the following data values: 5, 4, 6, 9, 3, 7, 7, 0, 9?

```
G       DC      0
L       DC      0
        READ    P
TOP     READ    N
        LOAD    N
        BE      STOP
        LOAD    P
        SUB     N
        BG      CONT
        LOAD    G
        ADD     =1
        STORE   G
        LOAD    N
        STORE   P
        BU      TOP
CONT    LOAD    L
        ADD     =1
        STORE   L
        LOAD    N
        STORE   P
        BU      TOP
STOP    LOAD    G
        MULT    =10
        ADD     L
        STORE   ANS
        PRINT   ANS
        END
```

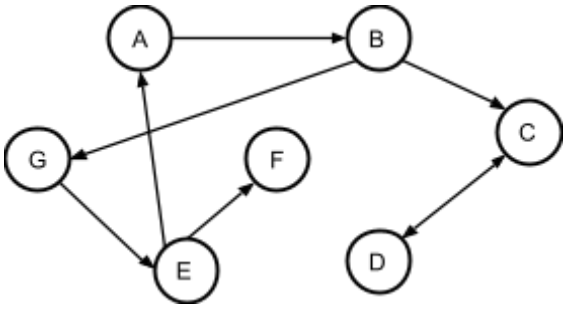
A. 62
B. 61
C. 42
D. 43
E. 25

American Computer Science League

2025-2026 ● Contest 4: Short Problems ● Senior Division

1. 图论

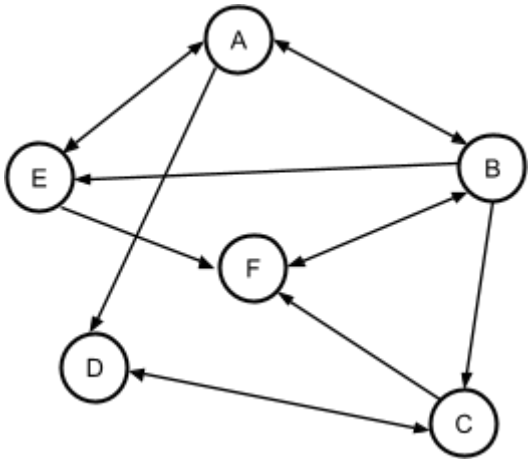
请问在以下有向图中，有多少个有序顶点对之间不存在长度为 2 的路径？示例，(B,A) 表示从 B 到 A 的路径。



A. 28
B. 39
C. 40
D. 41
E. 43

2. 图论

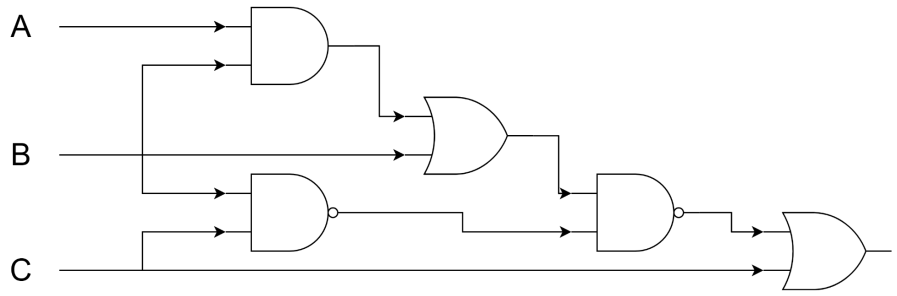
请问在这个有向图中，包含顶点 B 的环有多少个？



A. 4
B. 5
C. 6
D. 7
E. 8

3. 数字电路

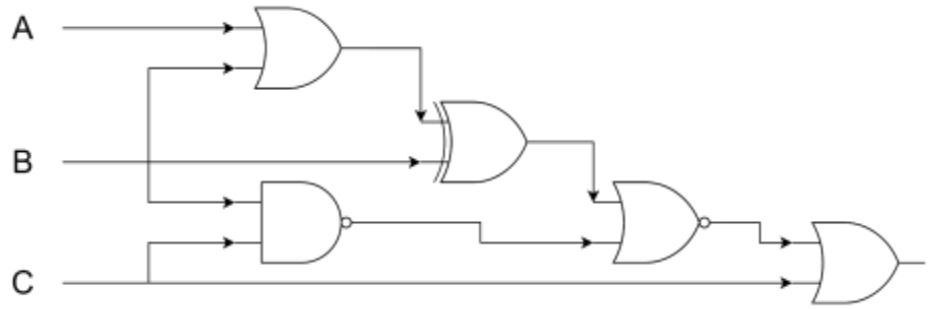
请问有多少个有序三元组能够使得以下数字电路为真？



A. 0
B. 2
C. 4
D. 6
E. 8

4. 数字电路

请使用尽可能少的运算，化简以下数字电路所表示的布尔代数表达式：



A. $B\overline{C}$
B. $BC + \overline{C}$
C. C
D. $B + C$
E. $B + \overline{C}$

5. 汇编语言

请问执行以下程序时会输出什么？

```
A       DC      3
B       DC      7
        LOAD    A
        MULT    B
        STORE   P
        DIV     =10
        STORE   C
        MULT    =-10
        ADD     P
        STORE   E
        ADD     C
        MULT    =10
        STORE   D
        LOAD    C
        MULT    =100
        ADD     D
        ADD     E
        STORE   ANS
        PRINT   ANS
        END
```

A. 231
B. 132
C. 321
D. 213
E. 123

6. 汇编语言

当使用以下数据值: 5, 4, 6, 9, 3, 7, 7, 0, 9 **执行该程序时，输出结果是什么**?

```
G       DC      0
L       DC      0
        READ    P
TOP     READ    N
        LOAD    N
        BE      STOP
        LOAD    P
        SUB     N
        BG      CONT
        LOAD    G
        ADD     =1
        STORE   G
        LOAD    N
        STORE   P
        BU      TOP
CONT    LOAD    L
        ADD     =1
        STORE   L
        LOAD    N
        STORE   P
        BU      TOP
STOP    LOAD    G
        MULT    =10
        ADD     L
        STORE   ANS
        PRINT   ANS
        END
```

A. 62
B. 61
C. 42
D. 43
E. 25